

LAB 1

Task 1.1: Superkey and Candidate Key Analysis

Relation A: Employee

Schema: Employee (EmpID, SSN, Email, Phone, Name, Department, Salary)

1. Six Different Superkeys:

{EmpID},{SSN},{Email},{EmpID, Name},{SSN, Email} ,{EmpID, SSN, Email, Phone, Name, Department, Salary}

2. Candidate Keys:

{EmpID},{SSN},{Email}

3. Primary Key Selection and Justification:

Selection: {EmpID}

Justification: EmpID is a surrogate key that is short, stable (numeric), and managed internally by the organization. SSN contains sensitive personally identifiable information (PII), making its use as a primary key/foreign key a security risk. Email can change if an employee updates their name or if the company domain changes.

4. Can two employees have the same phone number?

Answer: Based on the business rules and relation structure, the phone number is not defined as a unique identifier. Although the sample data shows unique phone numbers (555-0101, 555-0102, 555-0103), there is no UNIQUE constraint explicitly defined on the Phone attribute. Therefore, two employees could theoretically share the same phone number (e.g., a shared department phone).

Relation B: Course Registration

Schema: Registration (StudentID, CourseCode, Section, Semester, Year, Grade, Credits)

1. Minimum Attributes Needed for Primary Key:

{StudentID, CourseCode, Semester, Year}

2. Necessity of Each Attribute:

StudentID: Identifies the specific student enrolled.

CourseCode: Specifies which course the student is taking.

Semester and Year: Allow a student to retake the same course in different semesters/years (per Business Rule 1).

Note: Section is not required in the Primary Key because Business Rule 2 prohibits a student from registering for the same course section in the same semester.

3. Additional Candidate Keys:

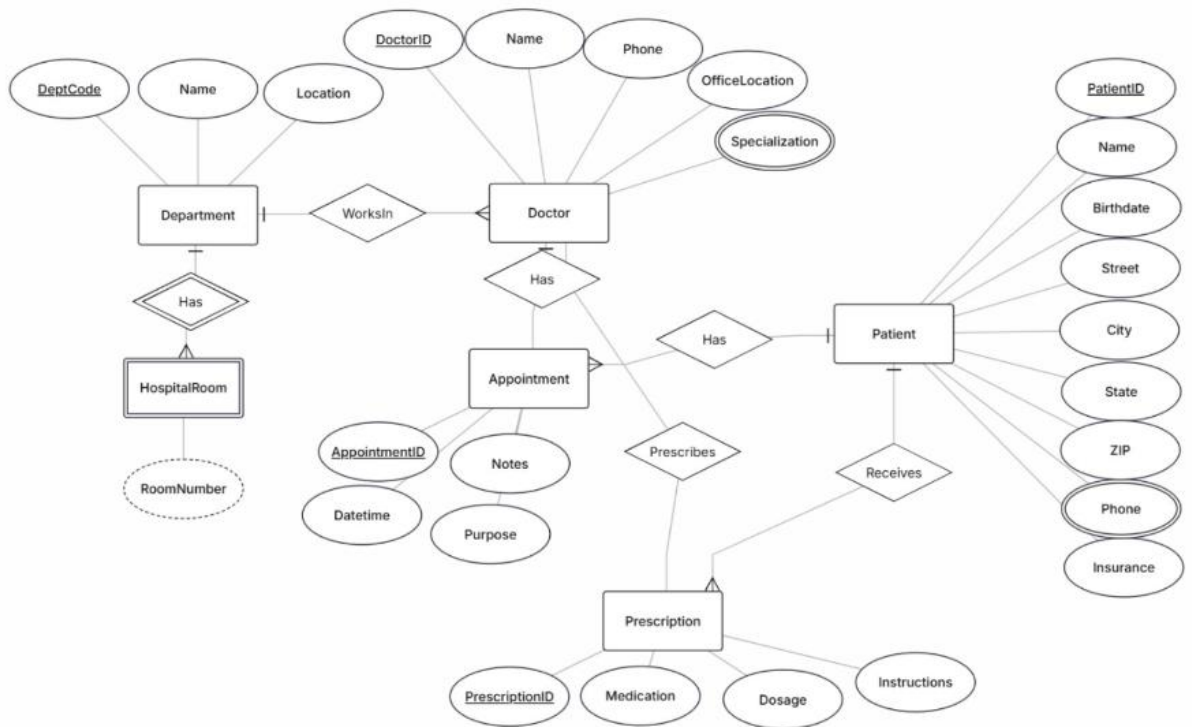
If sections are uniquely identified across all courses per semester: {StudentID, Section, Semester, Year}. Otherwise, no other candidate keys exist.

Task 1.2: Foreign Key Design

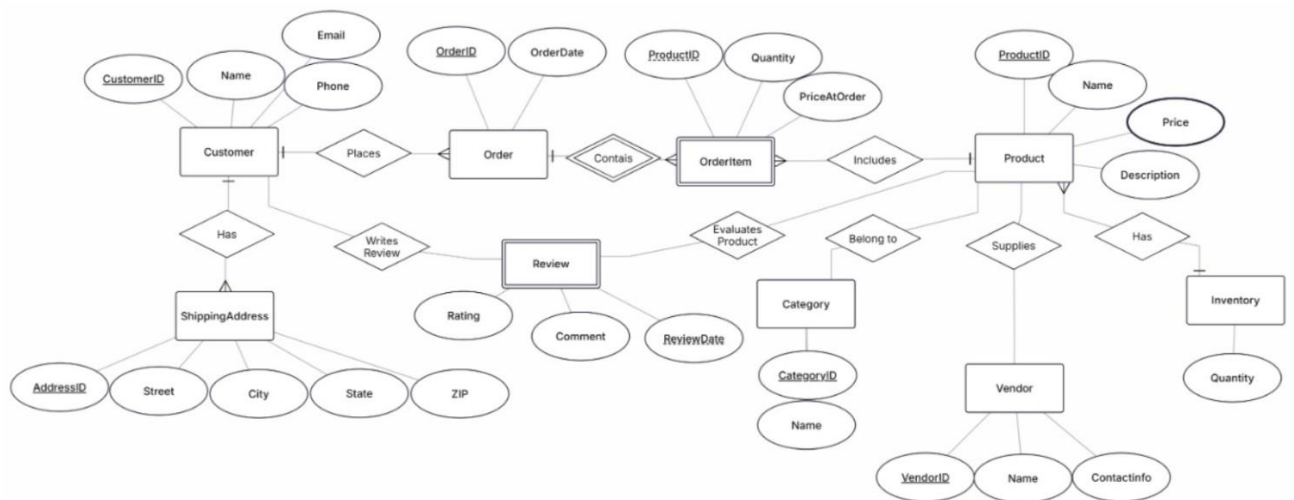
Foreign Key Relationships:

Child Table	Foreign Key Attribute	Parent Table	Referenced Primary Key	Description
Student	AdvisorID	Professor	ProfID	References the faculty advisor of the student
Course	DepartmentCode	Department	DeptCode	References the department offering the course
Department	ChairID	Professor	ProfID	References the professor who chairs the department
Enrollment	StudentID	Student	StudentID	References the enrolled student
Enrollment	CourseID	Course	CourseID	References the enrolled course

Task 2.1: Hospital Management System



Task 2.2: E-commerce Platform



Task 4.1: Denormalized Table Analysis

1. Functional Dependencies (FDs):

- 1) FD_1 : $StudentID \rightarrow StudentName, StudentMajor$
- 2) FD_2 : $ProjectID \rightarrow ProjectTitle, ProjectType, SupervisorID, StartDate, EndDate$
- 3) FD_3 : $SupervisorID \rightarrow SupervisorName, SupervisorDept$
- 4) FD_4 : $StudentID, ProjectID \rightarrow Role, HoursWorked$

2. Redundancies and Anomalies:

Redundancy: Student names, project details, and supervisor information are duplicated for every project assignment.

Insert Anomaly: Cannot record a new supervisor or project before assigning a student.

Delete Anomaly: Deleting the last student on a project deletes the project and supervisor details.

Update Anomaly: Updating a supervisor's department requires modifying multiple records.

3. 1NF Compliance:

Assuming all fields hold atomic values, the relation is already in 1NF.

4. 2NF Decomposition:

- Primary Key: {StudentID, ProjectID}
- Partial Dependencies: FD_1 and FD_2 depend on parts of the composite key.

2NF Relations:

1. Student (StudentID, StudentName, StudentMajor) — PK: StudentID
2. Project (ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, StartDate, EndDate) — PK: ProjectID
3. StudentProjectAssignment (StudentID, ProjectID, Role, HoursWorked) — PK: {StudentID, ProjectID}

5. 3NF Decomposition:

Transitive Dependency: In Project, $ProjectID \rightarrow SupervisorID \rightarrow SupervisorName, SupervisorDept$.

Final 3NF Schemas:

Student (StudentID, StudentName, StudentMajor)

Supervisor (SupervisorID, SupervisorName, SupervisorDept)

Project (ProjectID, ProjectTitle, ProjectType, SupervisorID, StartDate, EndDate)

FK: SupervisorID $\rightarrow$ Supervisor(SupervisorID)

StudentProjectAssignment (StudentID, ProjectID, Role, HoursWorked)

FK1: StudentID $\rightarrow$ Student(StudentID)

FK2: ProjectID $\rightarrow$ Project(ProjectID)

Task 4.2: Advanced Normalization

1. Primary Key:

Primary Key: {StudentID, TimeSlot}

Explanation: A student can attend only one course section in a given time slot, uniquely identifying the tuple.

2. Functional Dependencies:

1. $\text{StudentID} \rightarrow \text{StudentMajor}$
2. $\text{CourseID} \rightarrow \text{CourseName}$
3. $\text{InstructorID} \rightarrow \text{InstructorName}$
4. $\text{Room} \rightarrow \text{Building}$
5. $\text{CourseID, TimeSlot} \rightarrow \text{InstructorID, Room}$
6. $\text{InstructorID, TimeSlot} \rightarrow \text{Room, CourseID}$
7. $\text{Room, TimeSlot} \rightarrow \text{CourseID, InstructorID}$

3. BCNF Check:

The table is **NOT in BCNF** because functional dependencies exist where the determinant is not a superkey (e.g., $\text{StudentID} \rightarrow \text{StudentMajor}$).

4. BCNF Decomposition:

1. **Student** (StudentID, StudentMajor)
2. **Course** (CourseID, CourseName)
3. **Instructor** (InstructorID, InstructorName)
4. **RoomLocation** (Room, Building)
5. **CourseOffering** (CourseID, TimeSlot, InstructorID, Room)
6. **Enrollment** (StudentID, TimeSlot, CourseID)

Task 5.1: Real-World Application

